

Artificial Intelligence Ethics

MAIE 5103 · No. of Credits: 3

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Office Hours: *TBD*

Learn to recognize ethical risks before they become real-world harms.

Course Overview & Lecture 1: Introduction to AI Ethics

Welcome & Instructor Introduction

- A case-driven, discussion-based course on the ethics of AI.
- We investigate how AI systems are **designed, developed, and deployed** — and the societal impacts that follow.
- **About the instructor:** *Sirui Han (research & teaching interests to be confirmed).*
- **Contact:** siruihan@ust.hk · Office hours: TBD (announced via Canvas).
- Reach me by email; please allow ~2 business days for a reply.

This course connects technical design choices to their consequences for people and institutions.

Today's Agenda

- **Part 1 — Course Overview**
 - Objectives, weekly schedule, assessments, and course policies
 - **Break**
 - (*~10 min*)
- **Part 2 — Lecture 1: Introduction to AI Ethics**
 - What is AI? What is ethics? Why does it matter now? (*~50 min*)
- **Q&A — throughout and at the end**

We first establish how the course works, then build a practical vocabulary for ethical reasoning.

Course Description

This course delves into the ethical considerations involved in **designing, developing, and deploying** artificial intelligence (AI) systems. Students will investigate the societal impacts of AI and the ethical dilemmas that emerge in areas such as **privacy, bias, transparency, and accountability**. Through case studies and discussions, students will build a **framework for ethical decision-making** in the development and implementation of AI systems.

- **Key themes this term:** privacy & data protection · fairness & bias · accountability & transparency · safety & security · societal impact · AI policy & regulation.

Ethical analysis follows AI across research, design, deployment, use, and governance.

Why This Course Matters

- AI is now **pervasive and high-stakes** — it shapes credit, hiring, healthcare, criminal justice, and democracy.
- **Ethical harms are already happening:** biased decisions, privacy erosion, misinformation, surveillance.
- Ethics must be part of **design, not an afterthought**.
- **Over the term we examine real cases, e.g.:**
 - Recidivism risk scoring (COMPAS)
 - Public-service algorithms that profile the poor (Eubanks)
 - Facial recognition & surveillance; deepfakes; autonomous systems.

Automated decisions increasingly shape access to opportunities, services, rights, and public trust.

Course Intended Learning Outcomes (CILOs)

On successful completion of this course, students will be able to:

- **CILO-1** Acquire a comprehensive understanding of AI ethical considerations and social implications.
- **CILO-2** Develop own perspectives on the foundational ethical questions surrounding emerging technologies.
- **CILO-3** Critically analyze impacts and societal challenges of AI.
- **CILO-4** Apply knowledge to real-world AI-related issues and propose solutions.
- **CILO-5** Enhance written and oral communication skills to effectively discuss ideas related to AI ethics.

Mapping: *CILOs map onto assessments (e.g., Reading response → CILO 1, 2, 5; Final project → all) and onto weekly lectures.*

The goal is not only to know ethical concepts, but to evaluate, communicate, and act on them.

Course Structure: A 13-Week Journey

The course unfolds in four phases, building from foundations to a capstone project.

Phase	Weeks	Focus
Foundations	1 – 3	Introduction to AI Ethics · Privacy & Data Protection · Accountability, Responsibility & Transparency
Fairness & Bias	4 – 7	Fairness and Bias (Parts 1–2) · Beyond Bias (Parts 1–2) + Tutorials on Academic Writing & News Presentation
Applications & Governance	8 – 11	Ethics of Automation · Safety & Security · AI in Social Media · AI in Military & Politics · AI Policy & Regulations
Capstone	12 – 13	Final Project Presentations

Each phase builds toward an evidence-based capstone analysis of a real AI ethics problem.

Weekly Schedule

Order and contents may be adjusted according to teaching progress.

Week	Topic
1	Course Overview & Lecture 1: Introduction to AI Ethics
2	Lecture 2: Privacy and Data Protection
3	Lecture 3: Accountability, Responsibility, Transparency
4	Lecture 4: Fairness and Bias (Part 1) & Tutorial on Academic Writing
5	Lecture 4: Fairness and Bias (Part 2)
6	Lecture 5: Beyond Bias (Part 1) & Tutorial on News Presentation
7	Lecture 5: Beyond Bias (Part 2)
8	Lecture 6: The Ethics of Automation & Lecture 7: Safety and Security
9	Lecture 8: AI in Social Media
10	Lecture 9: AI in the Military & Lecture 10: AI in Politics
11	Lecture 11: AI Policy and Regulations
The semester moves from foundations to recurring issues, applied domains, and governance.	
12	Final Project Presentations

Assessments Overview

Criterion-referenced grading; letter grades assigned against descriptors (not curved).

Assessment Task	Weight	Due Date	Mapped CLOs
Participation	20%	In class	3, 4, 5
Quizzes (4 in total)	10%	In class	1, 2, 3
AI news presentation (Individual)	10%	~ End of October	2, 3, 4
Reading response (Individual)	15%	~ Mid-October	1, 2, 5
Case study analysis (Individual)	15%	~ Mid-November	3, 4, 5
Final project report (Group)	15%	~ Early December	1 – 5
Final project presentation (Group)	15%	In class (Wk 12–13)	1 – 5
Total	100%		

The assessment mix rewards preparation, analysis, communication, and integrated project work.

Assessment Details (1)

Participation · Quizzes · AI News Presentation · Reading Response

Participation (20%)

- Attendance + engaged discussion; bring the readings and contribute regularly.

Quizzes (10% — 4 in-class)

- Short, based on the weekly readings; held in Weeks 4, 7, 9, and 11.

AI News Presentation (10%, individual, ~ end-Oct)

- Present a significant recent AI news event (5–8 min); rubric on clarity, fact accuracy, and newsworthy analysis.

Reading Response (15%, individual, ~ mid-Oct)

- Critical written response to one reading from the Readings Pool; rubric on clarity, use of readings, critique, and citation.

Strong work is prepared, evidence-based, concise, and explicitly connected to ethical concepts.

Assessment Details (2)

Case Study Analysis · Final Project (Report + Presentation)

Case Study Analysis (15%, individual, ~ mid-Nov)

- Analyze a real-world AI ethics case from the Case Study Pool; rubric on clarity, source support & fact accuracy, introduction, analysis, citation.

Final Project — Report (15%, group, ~ early-Dec)

- In-depth group project on an AI ethics issue; rubric on clarity & organization, introduction, main body, accomplishments, recommendations, use of techniques/literature, citation.

Final Project — Presentation (15%, group, Wk 12–13)

- Present the project in class; rubric on demonstration, clarity, organization & timing, audience engagement, slides, and Q&A.

The case study diagnoses an existing problem; the final project develops and defends a response.

Required Texts & Weekly Readings

All available in HKUST library online resources or book collections.

- **Dignum, V.** *Responsible Artificial Intelligence: How to Develop and Use AI in a Responsible Way*. Springer, 2019.
- **Coeckelbergh, M.** *AI Ethics*. MIT Press, 2020.
- **Dubber, M. D., Pasquale, F. & Das, S. (Eds.).** *The Oxford Handbook of Ethics of AI*. Oxford University Press, 2020.
- **Eubanks, V.** *Automating Inequality: How High-Tech Tools Profile, Police, and Punish the Poor*. St. Martin's Press, 2018.
- **O'Neil, C.** *Weapons of Math Destruction: How Big Data Increases Inequality and Threatens Democracy*. Crown, 2016.

Every week, a list of readings will be provided and released on Canvas as additional resources.

The readings combine philosophy, responsible-AI practice, mathematical harms, and social impact.

Academic Integrity & Course AI Policy

Academic Integrity

- Uphold HKUST's Academic Honor Code; the University has zero tolerance for academic misconduct.
- Plagiarism = presenting others' work or ideas as your own without attribution. Refer to HKUST Academic Registry guidance.

Course AI Policy

- You may use generative AI as a **collaborator** (e.g., to aid searching), but all interactions must be credited in the assignment.

AI may support learning only when its use is permitted, disclosed, verified, and remains your own work.

Logistics, Communication & How to Succeed

- **Canvas is the hub:** readings, announcements, grades, and submissions.
- **Office hours:** TBD — email siruihan@ust.hk (~2 business days reply).
- **Late work:** not accepted without prior arrangement and supporting documents.
- **How to succeed:**
 - Do the weekly readings **before** class.
 - Speak up in discussions — participation counts **20%**.
 - Start the final project early; form your group early.
 - Build your own ethical voice over the term.

A reliable weekly rhythm is simple: read, contribute, reflect, and ask for help early.

PART B · LECTURE 1

Introduction to AI Ethics

What is AI, what is ethics, and why does it matter now?

To judge AI responsibly, we need to understand both the technology and the values it affects.

Lecture 1 Outline

Today's roadmap — seven sections.

01 What is AI? — definitions, history, how modern AI works

02 What is ethics? — ethics vs law; major ethical theories

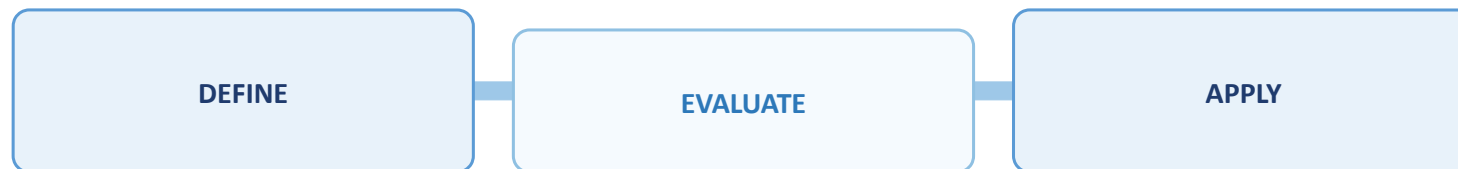
03 What is AI ethics & why now? — scope, urgency, actors

04 Key AI ethics issues — fairness, privacy, transparency, accountability, safety, impact

05 Case studies — criminal justice, public services, hiring, surveillance, deepfakes

06 Frameworks & decision-making — Responsible AI, principles, trade-offs, a method

07 Wrap-up — takeaways, reading, next lecture, Q&A



The lecture progresses from definition to evaluation, cases, and practical decision-making.

Lecture 1 Learning Objectives

By the end of this session, you should be able to...

DISTINGUISH

descriptive, normative, and technical questions

APPLY

multiple ethical lenses to the same case

MAP

stakeholders, values, benefits, and harms

JUSTIFY

a provisional decision and safeguards

Ethical competence means moving from recognition to reasoned action.

Opening Poll: Where Did AI Decide for You Today?

Think individually for 30 seconds, then compare with a partner.

MEDIA

FINANCE

EDUCATION

MOBILITY

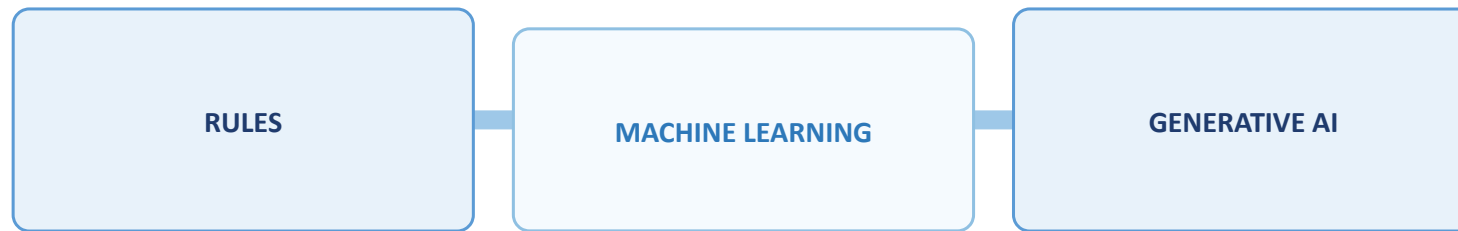
HEALTH

PUBLIC SERVICES

Name the decision, the data it may use, and one person who could be affected.

What is Artificial Intelligence?

- **Popular sense:** machines performing tasks that typically require human intelligence.
- **Technical sense (Russell & Norvig):** the study of rational agents that perceive and act.
- **Narrow (weak) AI:** task-specific systems — spam filters, face ID, translation.
- **General (strong) AI:** human-level ability across domains. *We are far from AGI today.*
- **Today's AI** ≈ pattern recognition from data at scale — not “thinking” in the human sense.

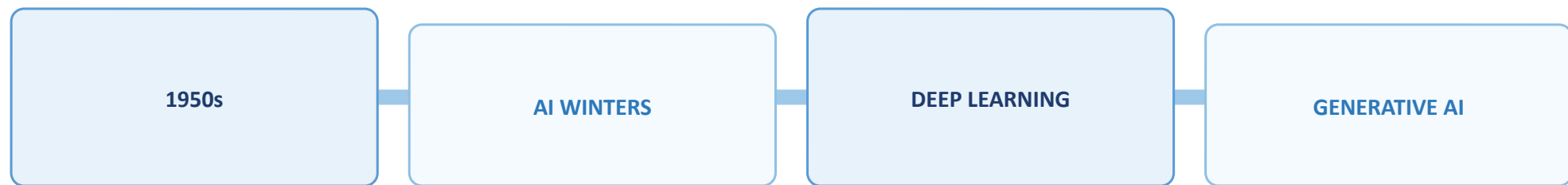


Most current AI learns patterns for specific tasks; fluent or capable behavior is not human understanding.

A Brief History of AI

AI is not new — but data + compute + methods made it explode.

- **1950** Turing Test — can a machine imitate human conversation?
- **1956** Dartmouth Workshop — the term “AI” is coined.
- **1970s–80s** Symbolic AI & expert systems, then two “AI winters.”
- **2000s** Machine learning rises as data and compute grow.
- **2012** Deep learning breakthrough (ImageNet).
- **2016** AlphaGo beats the world Go champion.
- **2020s** Generative AI & LLMs (GPT, diffusion) — fluent text, images, code.



AI history repeatedly cycles through optimism, technical limits, and renewed progress driven by new methods and resources.

AI History Is Also a History of Expectations

Capability changed—but so did the stories society told about AI.

1

1950s Foundational questions and symbolic ambition

2

1970s–90s AI winters expose limits in data, compute, and methods

3

2010s Deep learning scales pattern recognition

4

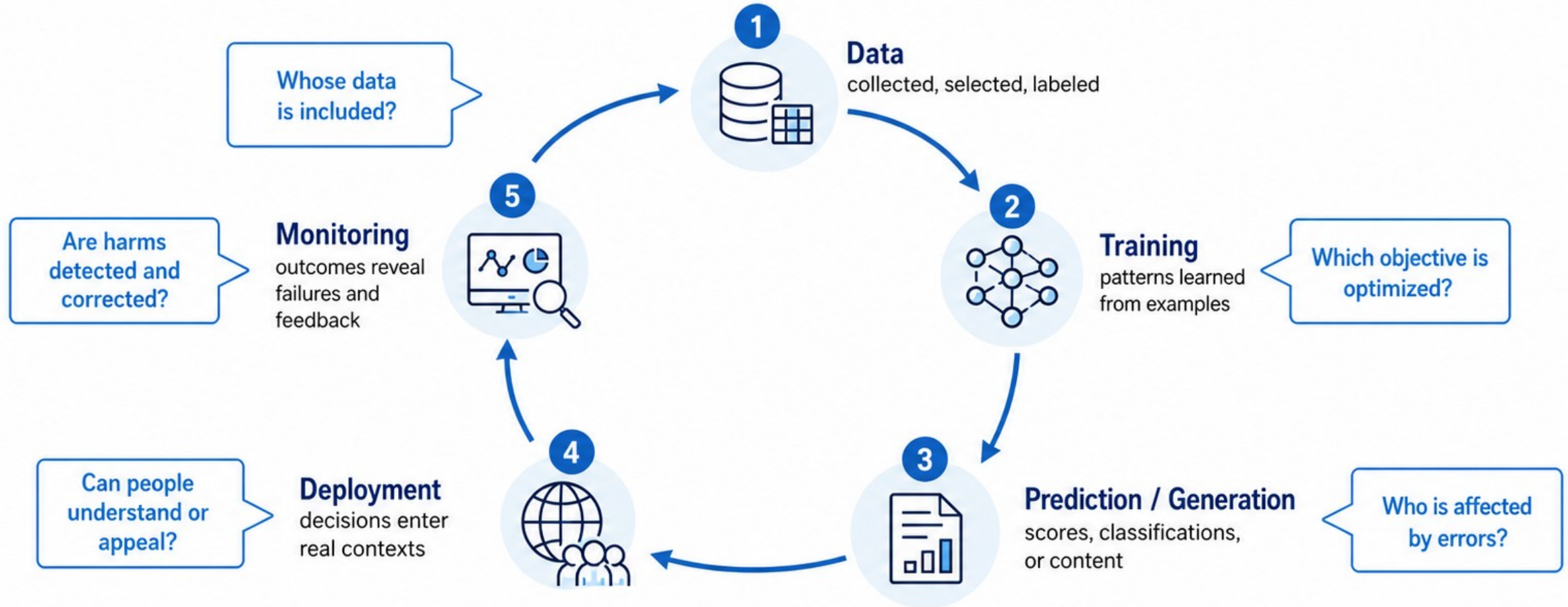
2020s Generative AI expands access—and expectations



Ethical analysis should separate demonstrated capability from hype, projection, and institutional incentives.

How Modern AI Works

A non-technical look at the loop — and why ethics issues arise.



Ethical choices enter at every stage — from data selection to real-world monitoring.

Generative AI & Large Language Models

What it is

- Trained on massive text; predicts the next token.
- Generates fluent text (ChatGPT), images (diffusion), code, audio.
- Emerging abilities: drafting, summarizing, translation, reasoning.

Limits relevant to ethics

- **Hallucination** — confident falsehoods.
- Non-determinism — different answers each run.
- Memorizes & may leak training data.
- No genuine understanding or grounding in the world.



Generative models can produce fluent outputs without knowing whether those outputs are true, fair, or appropriate.

LLMs: Fluency, Knowledge, and Verification Are Different

A confident sentence can be useful, false, unsupported—or all three.

FLUENCY

The output follows linguistic patterns and sounds coherent.

KNOWLEDGE

The model may reproduce useful associations, but it does not guarantee grounded understanding.

VERIFICATION

Claims still need evidence, provenance, context, and an accountable reviewer.

Treat fluency as a presentation property—not as proof of truth or understanding.

Mini Activity: Test a Generative-AI Answer

Read the answer, then audit it instead of merely reacting to it.

Sample answer: “This hiring model is fair because it does not use gender as an input.”

1

What factual evidence is offered?

2

Which assumptions are hidden?

3

Whose perspective is missing?

4

What would you verify before acting?

A responsible user turns an answer into a set of verification tasks.

Where AI is Deployed Today

AI is already in the infrastructure of daily life.

- **Search & recommendation** — what we see, read, and buy
- **Social media** — feeds, moderation, engagement
- **Finance & credit** — scoring, lending, insurance
- **Hiring & HR** — resumé screening, monitoring
- **Healthcare** — diagnosis, triage, allocation
- **Criminal justice** — risk scoring, policing
- **Autonomous systems** — vehicles, drones
- **Creative industries** — text, image, music generation



AI is often invisible infrastructure: it ranks, recommends, predicts, allocates, and moderates throughout daily life.

Capabilities, Limits & Uncertainties

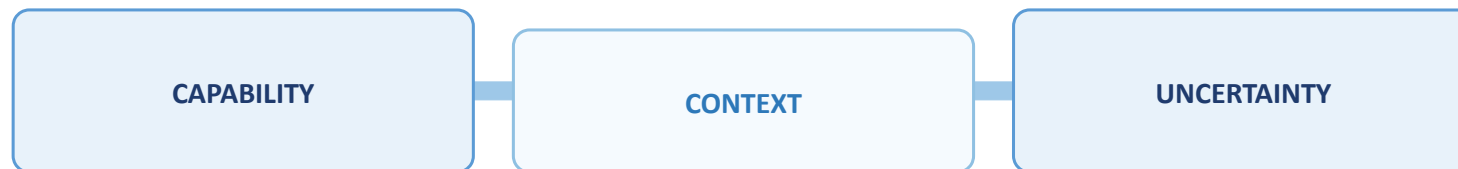
High confidence ≠ reliability.

Strengths

- Pattern recognition at vast scale.
- Speed and consistency beyond human capacity.
- Performance on well-defined, data-rich tasks.

Limits

- Brittleness — fails on distribution shift.
- Hallucination & lack of causal understanding.
- Non-reproducibility; vulnerable to adversarial inputs.



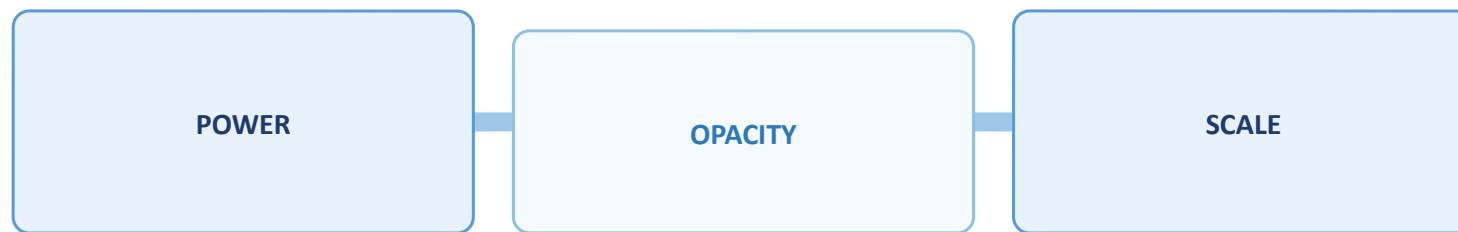
Benchmark performance is conditional; context change, uncertainty, and adversarial use can quickly undermine reliability.

From AI to AI Ethics

The bridge into Part 2.

Powerful + Opaque + Pervasive = ethics cannot be an afterthought.

- **Power:** high-stakes decisions over people's lives.
- **Opacity:** many models are black boxes — hard to inspect or explain.
 - **Pervasiveness:** billions are affected, often without knowing.



Powerful + opaque + pervasive systems create obligations to anticipate, explain, and remedy harm.

Break 1

Return question: Where in the AI lifecycle would you intervene first—and why?

15–20 MINUTES

What is Ethics?

- **Ethics:** systematic reflection on what is right/good and why.
- Morality = the beliefs and standards people actually hold; **ethics = the reasoned study of them.**
- Law = the enforceable minimum floor; ethics may demand more, or forbid what law allows.
- **Descriptive ethics:** what people believe.
- **Normative ethics:** what they ought to do.

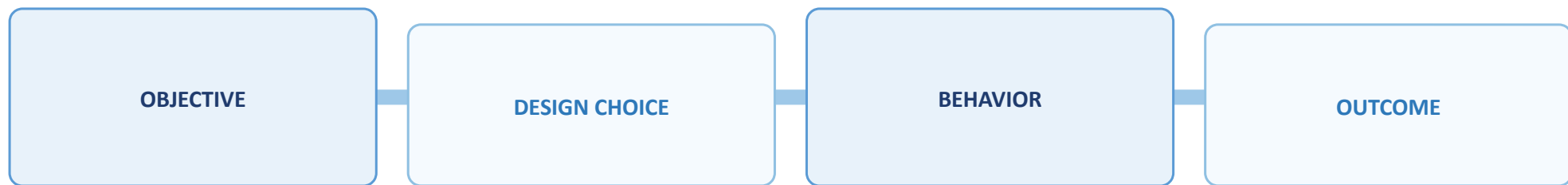


Law asks what is permitted; ethics asks what is justified and why, including when law is silent.

Why Ethics for Technology?

Technology is never neutral.

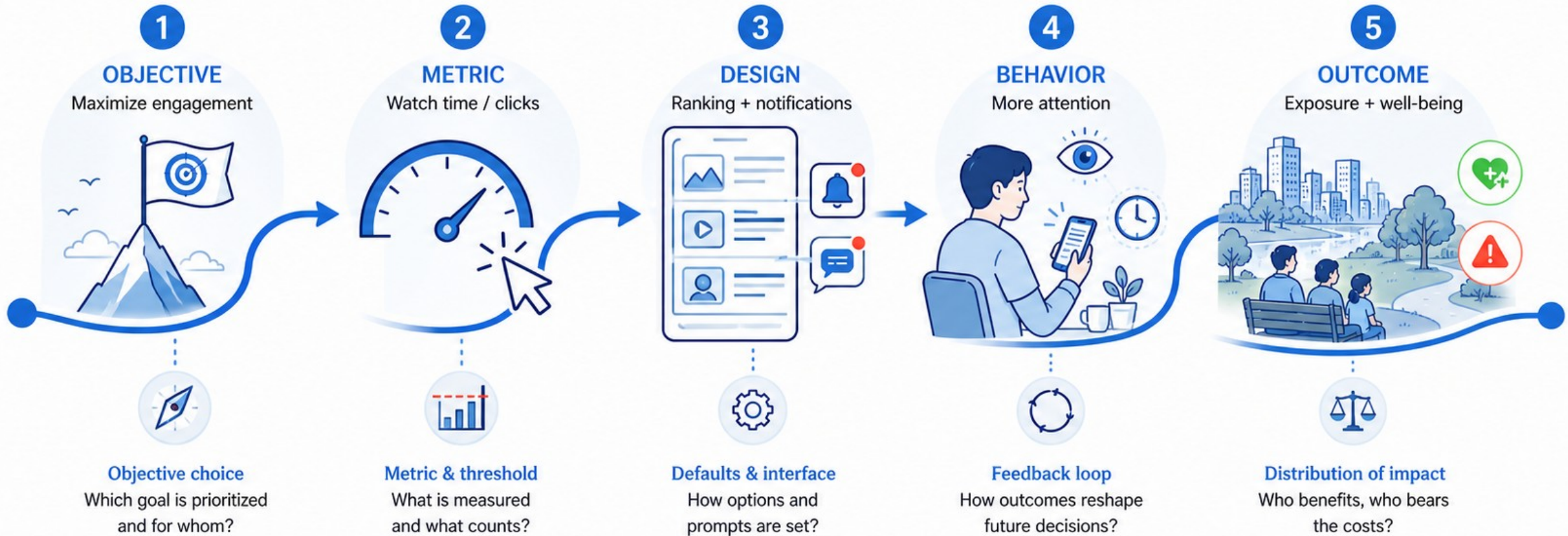
- “Tools shape how we perceive and act” — the **mediation thesis** (Coeckelbergh).
- Every design choice **embodies values** — what is optimized, whose interests count.
- A news feed optimizes engagement; a risk model optimizes a proxy for risk.
- So ethics must be **by design**, not bolted on after the harm.



Technology is not neutral because every objective, interface, and default privileges some values over others.

Values Become Technical Choices

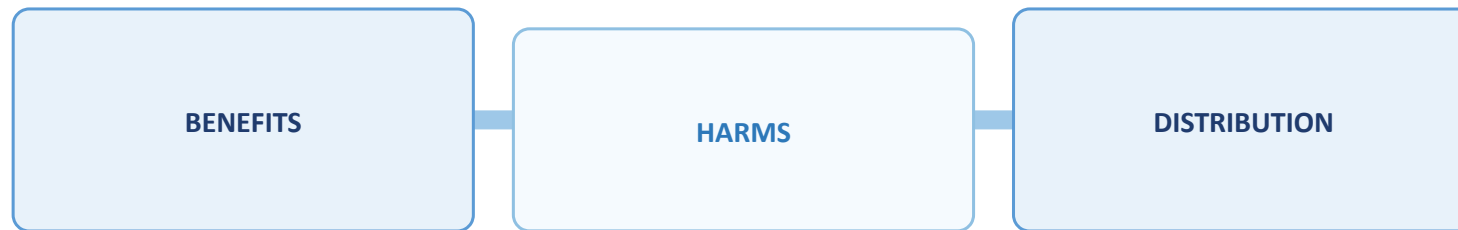
A recommendation system turns an abstract objective into a chain of concrete consequences.



Values enter through objectives, metrics, defaults, thresholds, and feedback loops.

Ethical Lens 1: Consequentialism / Utilitarianism

- Judge acts by their **consequences**; maximize overall well-being (Bentham, Mill).
- In AI: optimize a measurable objective — accuracy, profit, safety.
- **Strength:** pragmatic; fits cost–benefit reasoning.
- **Weakness:** “good” is hard to quantify; minorities can be sacrificed for the average.



Consequentialism asks whose benefits and harms count, over what time horizon, and under what uncertainty.

Ethical Lens 2: Deontology

- Some acts are right/wrong **regardless of outcomes** — grounded in duties, rights, consent.
- Kant: treat persons as **ends, not merely means**.
- In AI: consent, privacy, and human rights act as constraints on optimization.
- **Strength:** protects individuals and minorities.
- **Weakness:** rigid when values genuinely conflict.



A high-performing system can still be wrong if it violates rights, consent, dignity, or due process.

Ethical Lens 3: Virtue & Relational Ethics

- **Virtue ethics:** not “what act?” but what kind of agent/designer? (responsibility, humility, care).
- **Relational ethics (Coeckelbergh):** moral meaning arises in human–technology relations, not just abstract rules.
- Focus shifts to the character and relationships behind the technology.
- Especially apt for AI that mediates trust, care, and attention.



Virtue and relational ethics shift attention from isolated acts to character, care, dependence, and social context.

Other Lenses & When to Use Which

- **Justice (Rawls):** fair distribution; design to favor the least advantaged.
- **Care ethics:** relationships, vulnerability, dependence.
- **Rights-based:** human rights as side-constraints on design.
- **Principlism:** autonomy, beneficence, non-maleficence, justice.
- No single lens suffices — use several; they illuminate different facets *of a problem*.



Different lenses expose different blind spots; robust analysis usually needs more than one.

One Case, Four Ethical Lenses

Case: an AI hiring tool ranks applicants before any human review.

CONSEQUENCES

Do benefits outweigh harms across all groups?

DUTIES & RIGHTS

Are consent, dignity, and due process respected?

JUSTICE

Are opportunities and burdens distributed fairly?

CARE

Who is vulnerable or dependent on the decision?

The lenses do not merely produce different words; they direct attention to different evidence and obligations.

Small-Group Lens Exercise

Prepare a one-minute recommendation using only your assigned lens.

1

Identify the key ethical question

2

Request the evidence your lens needs

3

Recommend deploy, redesign, audit, or stop

4

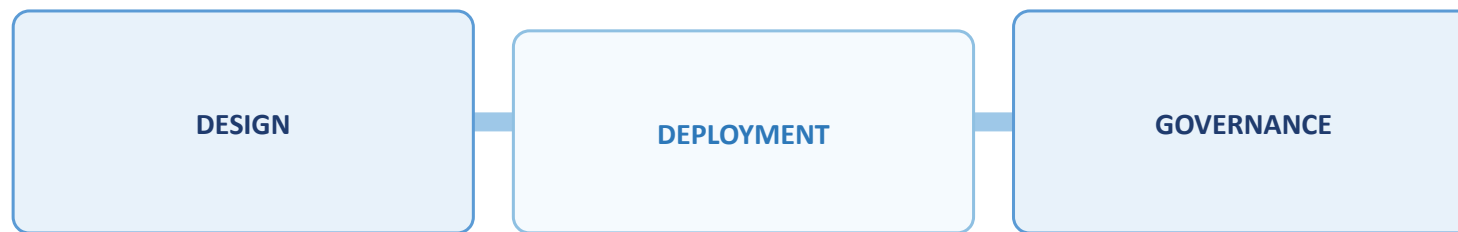
Name one weakness in your own analysis

Report back with a claim, the evidence required, and a limitation—not a slogan.

What is AI Ethics?

Definition & scope.

- **AI ethics:** the critical study of moral issues arising in AI research, design, deployment, and governance (Dignum; Oxford Handbook).
- **Adjacent fields overlap but differ:** data ethics, robot ethics, computer ethics.
- **Four levels:** research integrity → responsible design → deployment context → governance.



AI ethics studies moral questions across the full socio-technical lifecycle, not only model behavior.

What Counts as an “AI Ethics Issue”?

Is (descriptive)

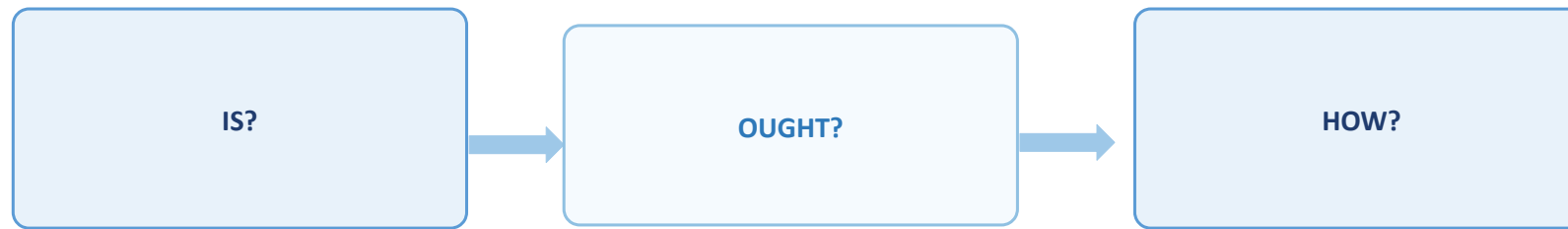
- *Does this model discriminate?*

Ought (normative)

- *Should we deploy it?*

How (technical)

- *What safeguards & redress?*



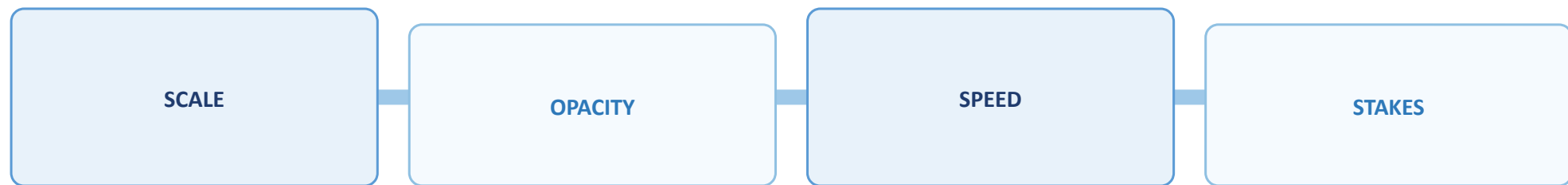
AI ethics spans all three — description, judgment, and method.

A complete analysis combines descriptive evidence, normative judgment, and practical safeguards.

Why AI Ethics, Why Now?

Four structural reasons.

- **Scale:** billions affected, often simultaneously.
- **Opacity:** models are black boxes — hard to inspect or explain.
- **Speed:** rapid, iterative deployment outpaces reflection.
- **Stakes:** decisions over credit, health, liberty, and democracy.
- *Plus: rising public concern and the first wave of regulation.*

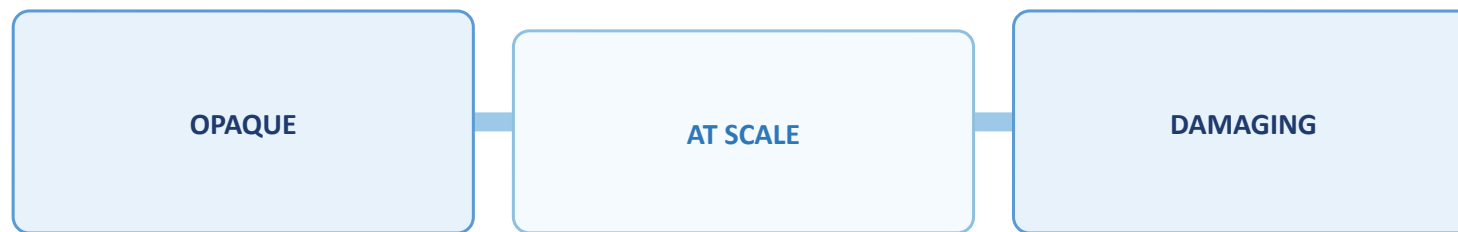


Scale, opacity, speed, and high stakes make after-the-fact review too slow and too costly.

O'Neil's "Weapons of Math Destruction"

A working framework for harmful algorithms (O'Neil, 2016).

- A WMD = Opaque + at Scale + Damaging.
- **Opacity:** the affected cannot see how decisions are made.
- **Scale:** applied to many people at once.
- **Damage:** errors cause real, often irreversible harm.
- **Proxies:** models substitute stand-ins (e.g., zip code for risk) for what they can't measure.
- **Feedback loops:** predictions drive actions that become new data, reinforcing the model.

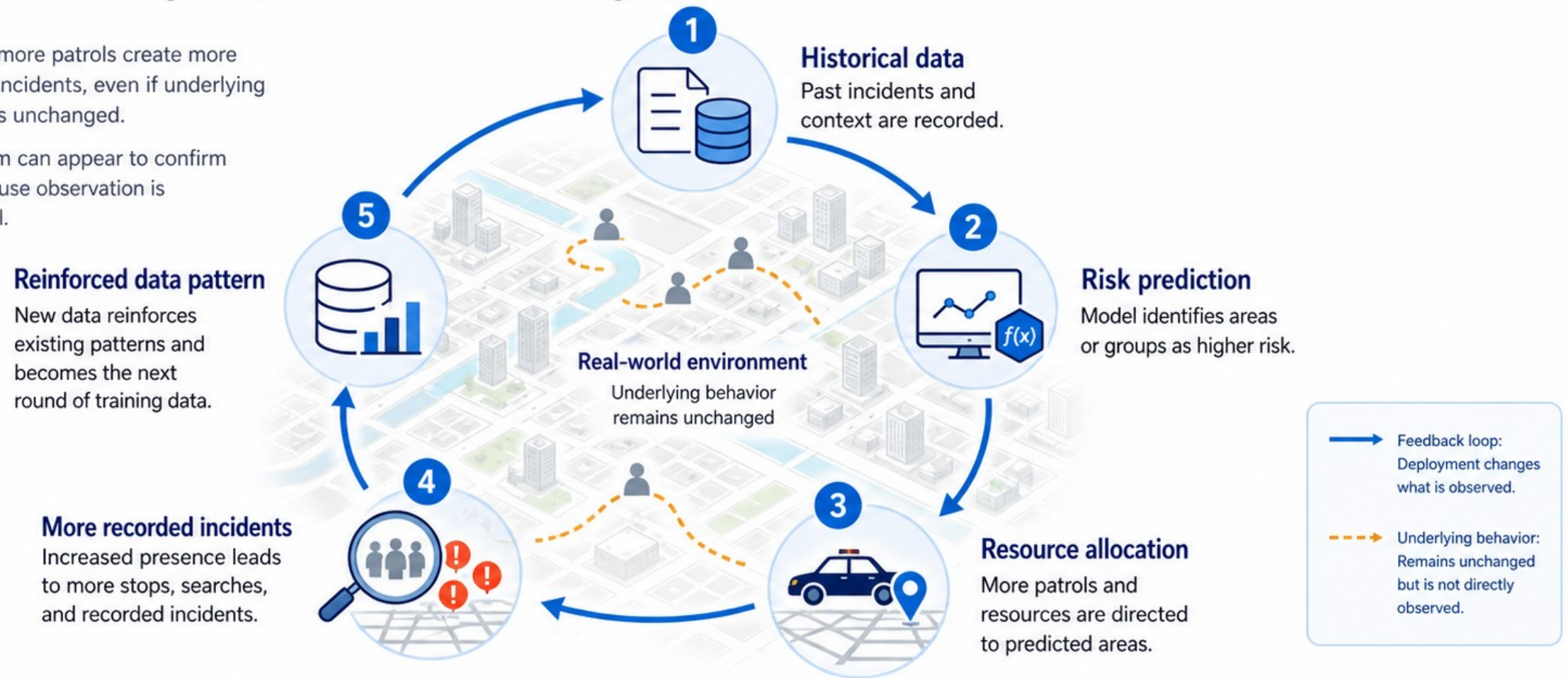


Opaque, scaled, damaging systems often reinforce themselves through feedback loops.

How Harmful Algorithms Create Feedback Loops

Predictions can change the world that later becomes training data.

- Example: more patrols create more recorded incidents, even if underlying behavior is unchanged.
- The system can appear to confirm itself because observation is not neutral.



Monitoring must examine how deployment changes data—not only whether the model matches past labels.

The AI Ethics Landscape

A growing ecosystem of actors.

- **Researchers & academics:** concepts, methods, critique.
- **Industry:** AI-ethics and AI-safety teams; responsible-AI programs.
- **Governments & regulators:** emerging rules and risk-based frameworks.
- **Civil society:** advocacy, audits, affected communities.
- **Standards bodies:** OECD, UNESCO, IEEE — principles and standards.
- *From principles → slowly → enforceable practice.*

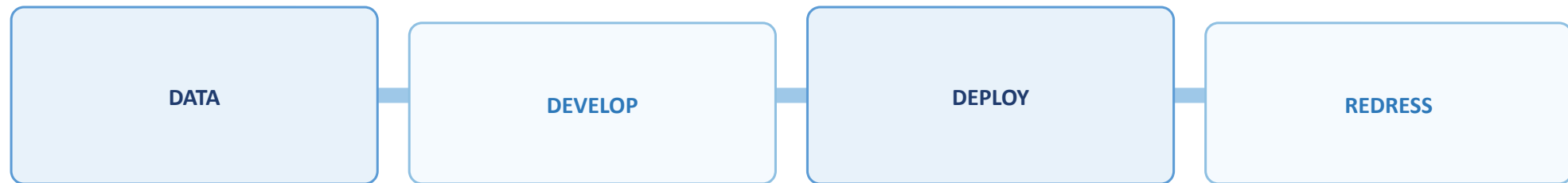


No single actor controls the full lifecycle, so effective governance must coordinate multiple forms of expertise and authority.

Stakeholders & Responsibilities

The “many hands” problem (Dignum).

- **The pipeline:** data providers → developers → deployers → users → affected communities → regulators.
- Responsibility is distributed — and **gaps enable evasion** (“no one is responsible”).
- **Dignum’s framing:** Responsible AI = who is responsible + what we design + how we govern.

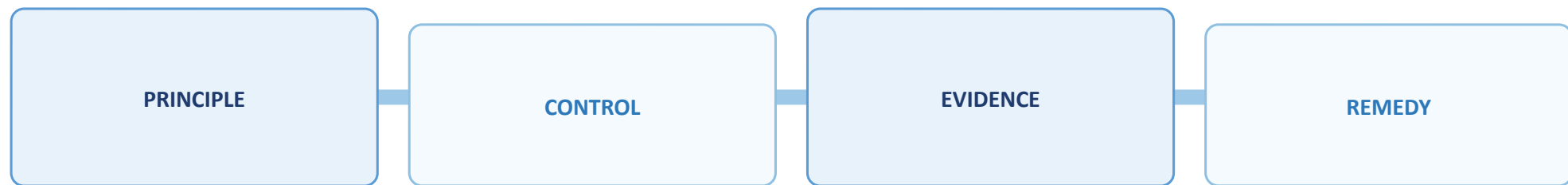


Distributed work does not eliminate responsibility; it makes ownership, handoffs, and escalation more important.

From Principles to Practice

The implementation gap.

- Hundreds of AI-ethics frameworks; recurring principles (fairness, transparency, accountability).
- **But:** few are enforceable; hard to operationalize; risk of “ethics washing”.
- **The gap: principles → practice → governance.**
- *Bridging it is the work of the rest of this course (preview: Week 11 — AI Policy & Regulations).*



Principles become credible only when translated into controls, evidence, monitoring, appeal, and remedy.

Map of Key AI Ethics Issues

Six recurring clusters — each deep-dived in later weeks.

- **Fairness & bias** (Wk 4–7)
- **Privacy & data protection** (Wk 2)
- **Transparency & explainability** (throughout)
- **Accountability & responsibility** (Wk 3)
- **Safety, security & reliability** (Wk 8)
- **Broader societal impact** (Wk 9–11)

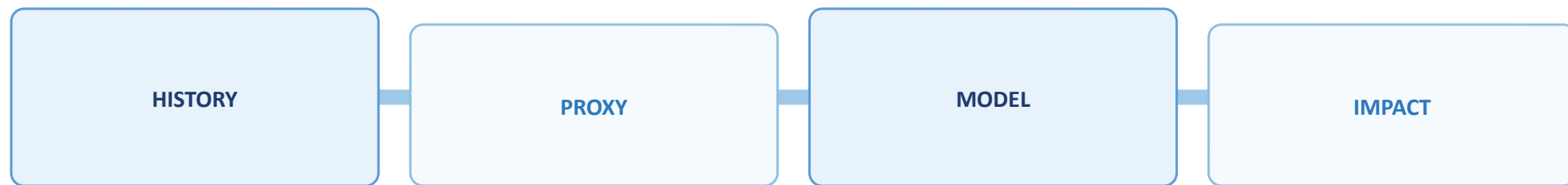


Fairness, privacy, transparency, accountability, safety, and societal impact are deeply interconnected.

Fairness & Bias

An introduction — deep dive in Weeks 4–6.

- Bias $\neq$ just accuracy gaps; it is unjustified differential harm.
- **Where bias enters:**
 - historical data · label noise · feature / proxy selection · model choices · feedback loops.
- **Teaser cases:** hiring, facial recognition, credit (Weeks 5–7).

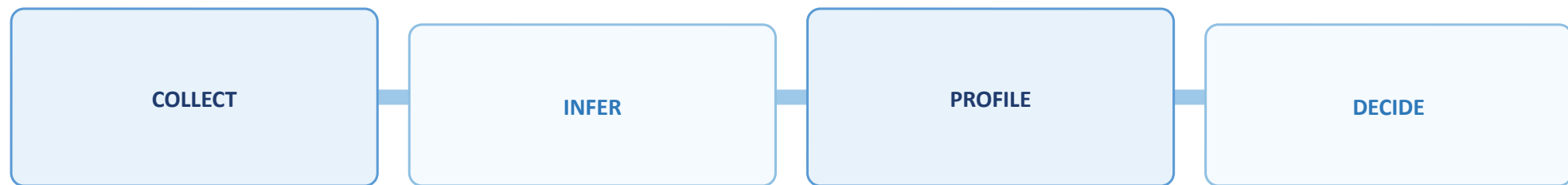


Bias is not only an accuracy gap; it is an unjustified pattern of differential burden or harm.

Privacy & Data Protection

An introduction — deep dive in Week 2.

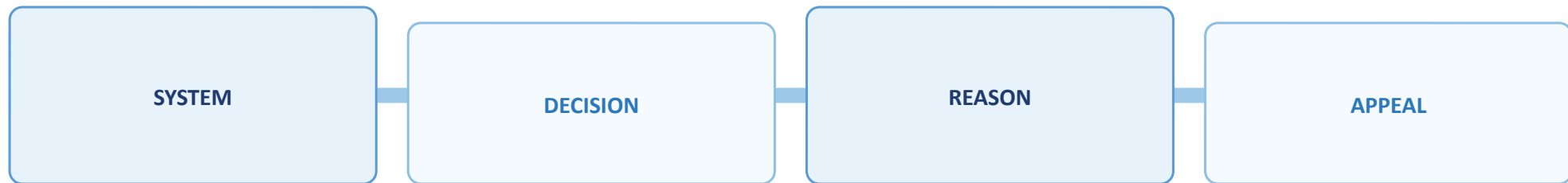
- AI infers sensitive attributes from “non-sensitive” data.
- **Re-identification:** “anonymized” records can be matched back to people.
- Mass surveillance chills behavior and dissent.
- **Core tension:** utility vs privacy.



Seemingly harmless data can reveal identity, health, beliefs, location, and vulnerability through inference.

Transparency & Explainability

- **The question:** “Why did the model decide X about me?”
- Many models are **black boxes** — hard to inspect or explain.
- Interpretable models vs post-hoc explanations (which can mislead).
- **Trade-off:** interpretability $\leftrightarrow$ accuracy / proprietary secrecy.
- A “right to an explanation” (GDPR-style) is emerging.



Useful explanation depends on the audience: developers, regulators, deployers, and affected people need different answers.

Explainability Depends on Who Is Asking

Different audiences need different forms of transparency.

DEVELOPERS

failure modes, features, data, tests

DEPLOYERS

appropriate use, thresholds, monitoring

REGULATORS

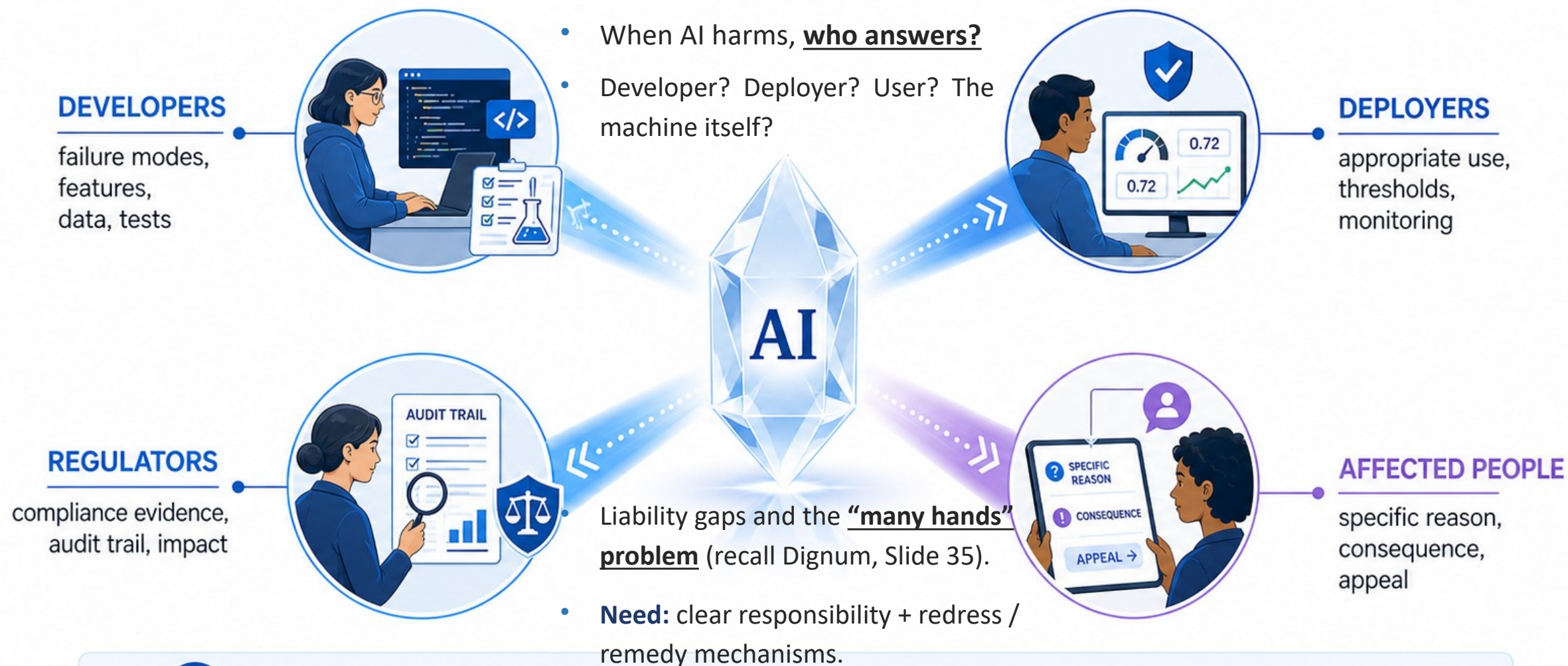
compliance evidence, audit trail, impact

AFFECTED PEOPLE

specific reason, consequence, appeal

An explanation is useful only when it supports the audience's next legitimate action.

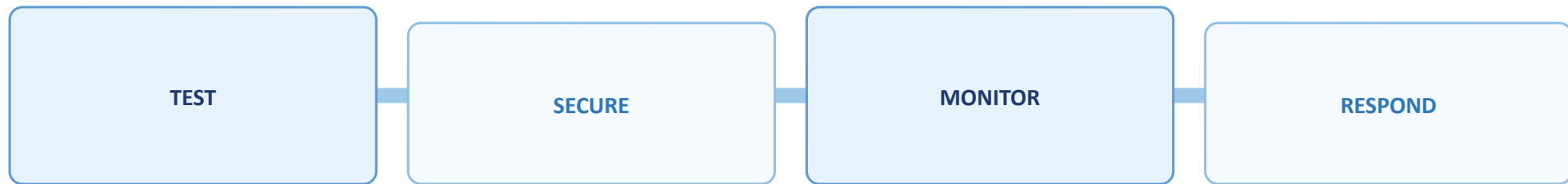
Explainability Depends on Who Is Asking



An explanation is useful only when it supports the audience's next legitimate action.

Safety, Security & Reliability

- **Reliability:** fails safely; robust to distribution shift.
- **Security:** adversarial attacks, model extraction, data poisoning.
- **Dual use:** the same tech can heal or harm (e.g., bio-research vs bio-risk; deepfakes).
- High confidence $\neq$ **reliability** — *test before you trust*.



Safety and security require testing before deployment and monitoring under real-world change after deployment.

Broader Societal Impacts

- **Labor:** automation & displacement; shifting skill demands.
- **Inequality & power:** benefits concentrate; the poor are over-profiled (O'Neil, Eubanks).
- **Democracy:** misinformation, polarization, manipulation.
- **Environment:** energy & water footprint of large models.

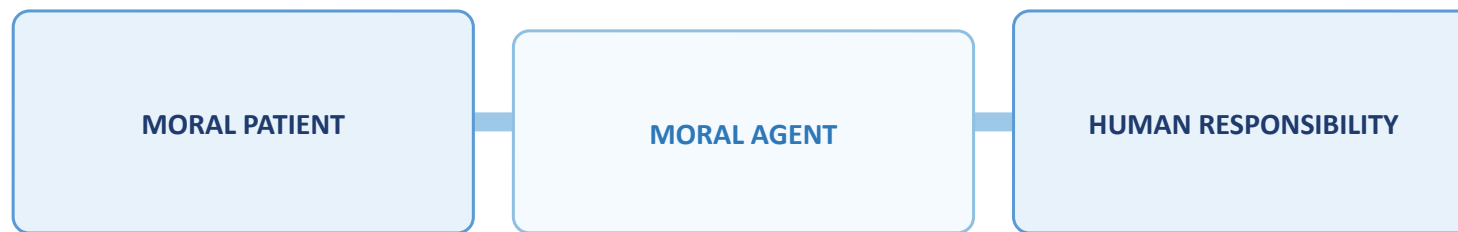


Aggregate benefits can coexist with concentrated harms, unequal power, and long-term institutional change.

Moral Status & AI Agency

A philosophical question (Coeckelbergh).

- **Moral patient:** should we care about AI itself? (suffering? rights?)
- **Moral agent:** can AI be held responsible? Most say no — no intent or consciousness.
- **Coeckelbergh's relational view:** moral status depends on relations & contexts, not fixed essence.
- **Practical focus:** *humans affected by AI bear the primary moral weight.*



Moral status asks whether AI can be harmed; moral agency asks whether it can be responsible — these are different questions.

Debate: Can an AI System Be Morally Responsible?

Separate autonomy in operation from responsibility in the moral and legal sense.

ARGUMENT FOR

Advanced systems can select actions, adapt, and operate beyond immediate human control. Responsibility concepts may need to evolve.

ARGUMENT AGAINST

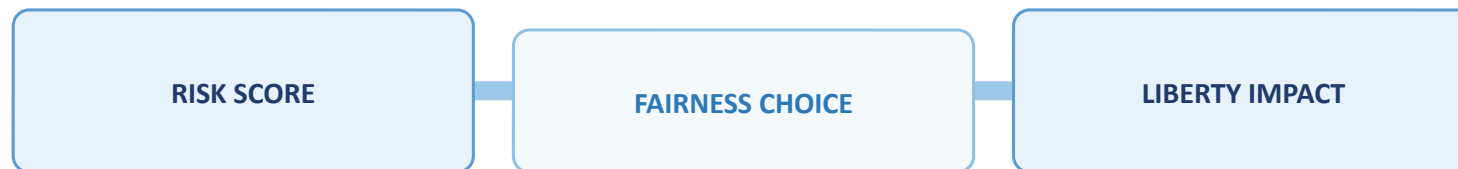
Systems lack human intentions, social membership, and moral understanding. Assigning responsibility to AI may let institutions evade accountability.

Even if agency remains contested, human institutions must still assign answerability and remedy.

Case 1: Algorithmic Bias in Criminal Justice

COMPAS — recidivism risk scoring (O’Neil).

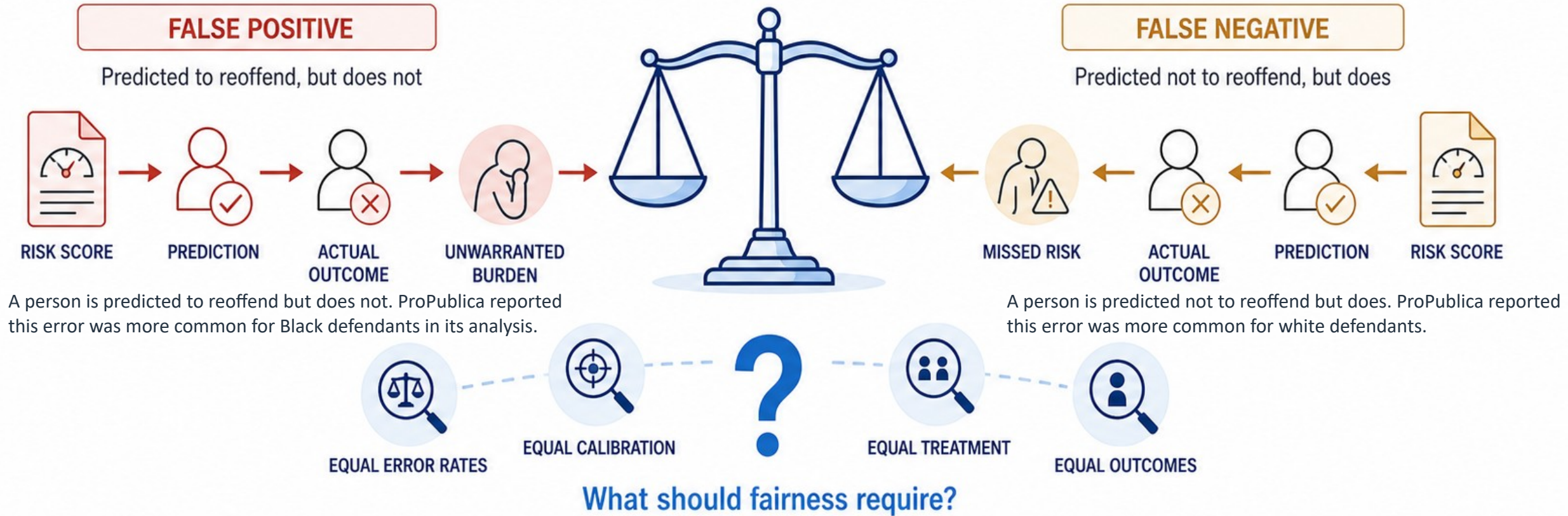
- **COMPAS:** a recidivism-risk tool used in US sentencing and bail decisions.
- **ProPublica (2016):** over-predicted recidivism for Black defendants; under-predicted for white.
- **Tensions:** accuracy vs fairness; opacity; stakes = liberty.
- **Lesson:** a “neutral” score encodes bias; criminal history acts as a proxy for race.



The COMPAS debate shows that competing fairness definitions can produce different judgments from the same scores.

COMPAS: Accuracy and Fairness Can Conflict

The same scores can be evaluated using different—and sometimes incompatible—fairness criteria.



Discussion: Should fairness mean equal error rates, equal calibration, equal treatment, or equal outcomes?



A fairness metric encodes a normative choice about which errors and inequalities matter.

Break 2

Return question: Which AI ethics issue is hardest to govern—and what makes it difficult?

15–20 MINUTES

Case 2: Scoring the Poor

The “digital poorhouse” (Eubanks, 2018).

- Three systems that **profile, police, and punish the poor**:
 - IN welfare-eligibility automation — benefits denied en masse.
 - Allegheny County child-welfare predictive-risk model.
 - LA coordinated-entry system for homelessness triage.
- **Patterns:** proxies for poverty; little transparency or recourse; automation of austerity.
- **Lesson:** the most vulnerable bear the most opaque, punitive AI.



Automation can scale administrative burden and denial when notice, human review, and appeal are weak.

Case 3: WMDs in Hiring, Credit & Insurance

Three O'Neil cases.

Hiring

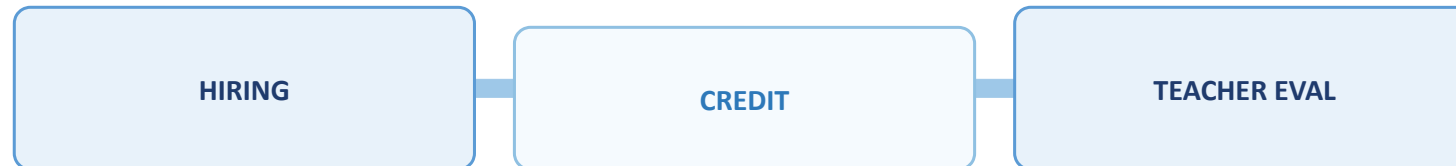
- Résumé-screening models learn biased proxies (e.g., gendered terms).
- *Opaque + scaled + damaging.*

Credit & insurance

- Zip code as a proxy for race / risk; opaque pricing.
- *Opaque + scaled + damaging.*

Teacher eval

- Value-Added Models punish teachers for factors beyond their control.
- *Opaque + scaled + damaging.*



Proxy variables and feedback loops allow discrimination to persist even when protected attributes are removed.

Case 4: Facial Recognition & Surveillance

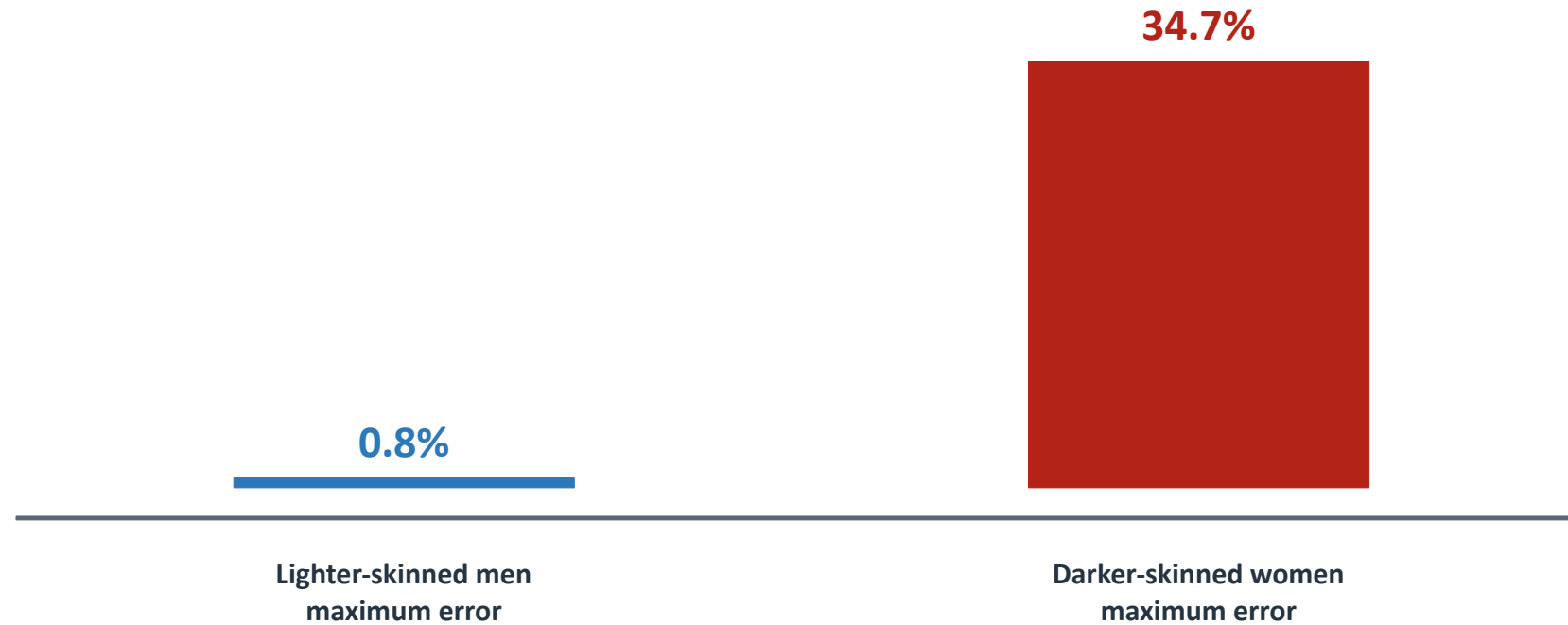
- Buolamwini & Gebru (2018), Gender Shades: error far higher for darker-skinned women than lighter-skinned men.
- **Uses:** unlock, tagging, access control.
- **Abuses:** mass surveillance, wrongful arrest.
- **Harms:** misidentification → wrongful arrest; chilling effect.
- **Need:** accuracy equity, consent, and limits on use.



Accuracy alone cannot justify surveillance; legitimacy also depends on purpose, consent, proportionality, and oversight.

Gender Shades: Intersectional Evaluation Changes the Result

Aggregate accuracy can hide severe subgroup disparities.

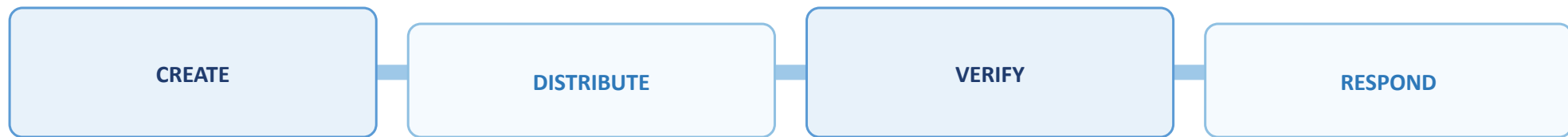


Reported maxima across the commercial systems evaluated in the 2018 study

Disaggregate performance by intersecting groups before concluding that a system is accurate or fair.

Case 5: Deepfakes, Misinformation & Generative AI

- Cheap, convincing synthetic audio / video / images.
- **Harms:** non-consensual intimate imagery; election disinformation; erosion of trust (“the liar’s dividend”).
- Generative AI lowers the cost of fakes; detection is an arms race.
- **Need:** provenance / watermarking, media literacy, platform policy.

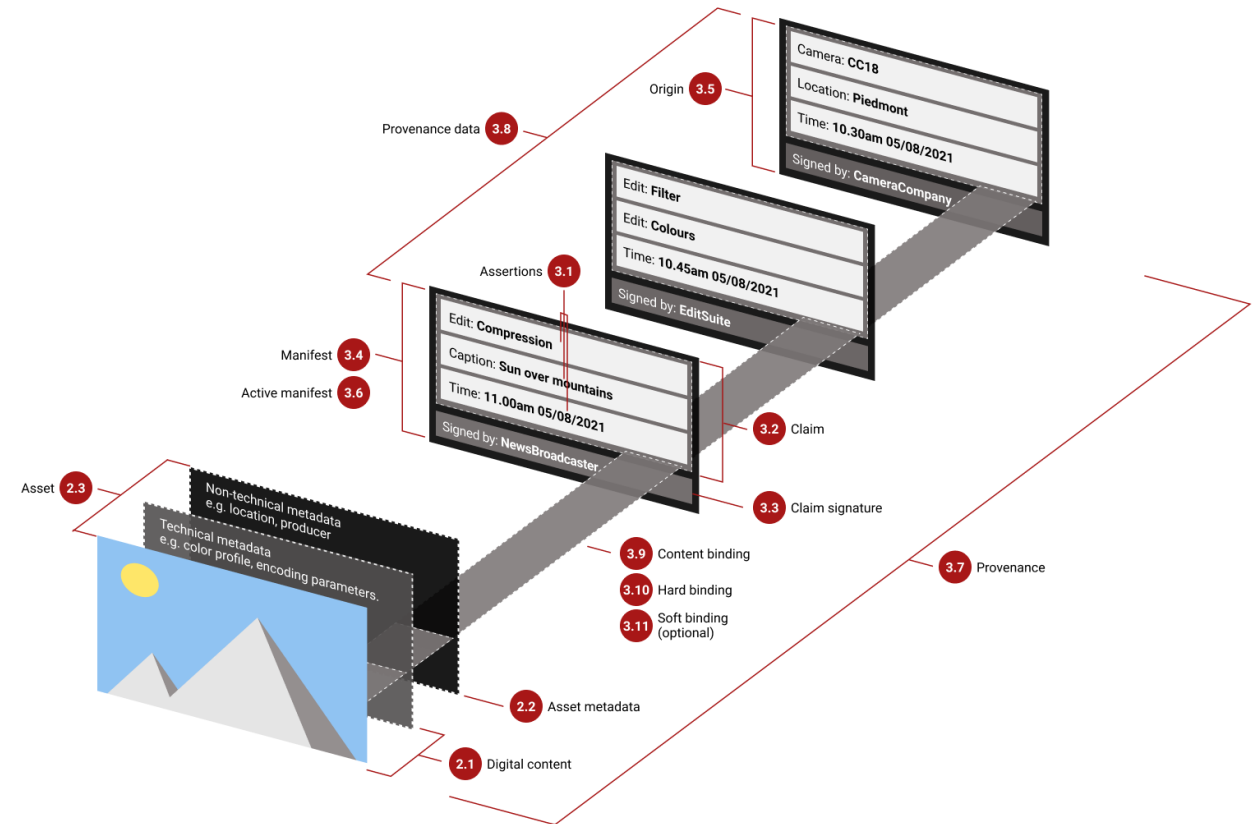


Deepfake governance needs layered responses: provenance, detection, platform policy, law, literacy, and victim support.

Content Provenance: What C2PA Can—and Cannot—Prove

Signed provenance can record origin and edits, but it does not certify truth.

- A manifest can bind claims and signatures to an asset.
- Edit history helps users inspect provenance and detect tampering.
- Authentic provenance can still accompany misleading content.



Provenance supports verification; it does not replace contextual judgment or fact-checking.

Case 6: Autonomous Systems — Vehicles & Weapons

- **Autonomous vehicles:** split-second harm; the Moral Machine experiment shows cross-cultural variation.
- **Lethal autonomous weapons (LAWS):** machines that select targets.
- **Responsibility gap:** who is liable / accountable for an autonomous decision?
- **Stakes:** *life-and-death decisions delegated to opaque systems.*



Vehicles and weapons differ in purpose, but both raise questions about meaningful human control and accountability.

Structured Debate: Meaningful Human Control

Compare autonomous vehicles and lethal autonomous weapons without assuming they are ethically equivalent.

	AUTONOMOUS VEHICLES	AUTONOMOUS WEAPONS
PURPOSE	mobility and safety	selection and use of force
TIME PRESSURE	collision avoidance	targeting and engagement
HUMAN ROLE	design, supervision, takeover	authorization, judgment, abort
FAILURE IMPACT	injury and public safety	life, conflict, international law
ACCOUNTABILITY	manufacturer, operator, regulator	commanders, states, developers

Meaningful control requires timely information, genuine authority, and an identifiable responsible actor.

“Responsible AI” — the Triad

Dignum: Responsible AI is socio-technical.

- **Responsible AI = Responsibility + Design + Governance.**
- **Who is responsible:** agency & accountability across the pipeline.
- **What we design:** values-by-design — ethics built into the system.
- **How we govern:** oversight, auditing, and redress.
- Integrates ethics across the whole lifecycle, *not just the code*.

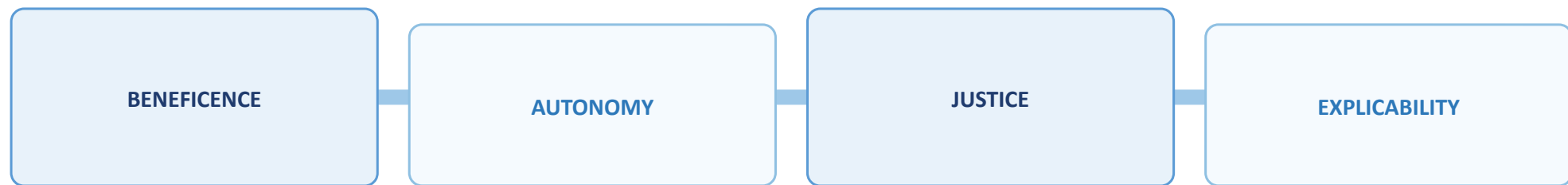


Responsible AI links who is accountable, what is designed, and how governance verifies performance in practice.

Ethical Principles for AI

Many frameworks converge on a core set (Floridi et al.).

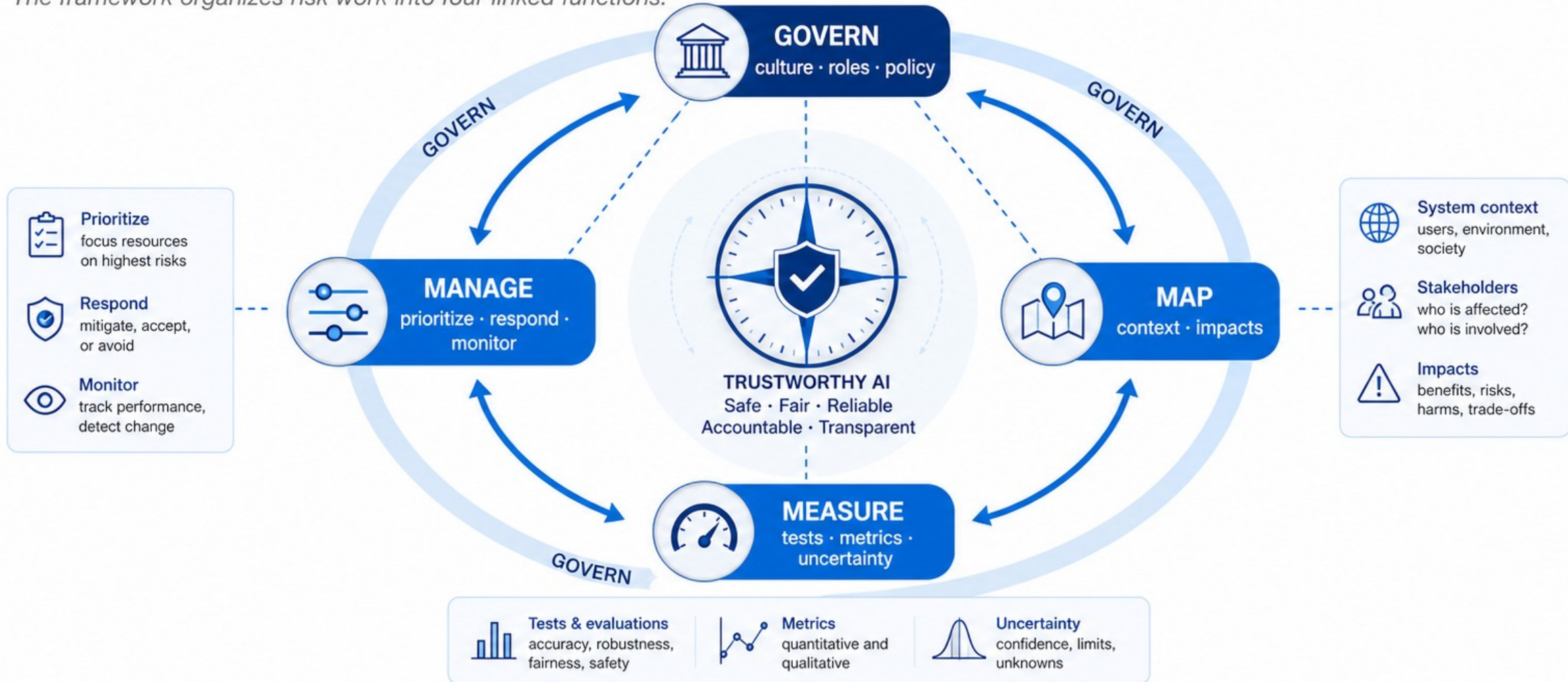
- **Beneficence:** do good — promote well-being.
- **Non-maleficence:** do no harm.
- **Autonomy:** preserve human agency and choice.
- **Justice:** fair and equitable.
- **Explicability:** intelligible and accountable.
- *Also: EU Ethics Guidelines, UNESCO, OECD AI Principles.*



Ethical principles work best as review questions that guide design evidence, not as slogans.

From Principles to Risk Management: NIST AI RMF

The framework organizes risk work into four linked functions.



Risk management is continuous: context, measurement, response, and governance must inform one another.

From Global Values to Policy Action: UNESCO

Human rights and dignity connect ethical principles to public policy.

CORE VALUES

-  Human rights & dignity
-  Peaceful, just societies
-  Environment & ecosystems
-  Diversity & inclusion



ACTION AREAS

-  Data governance
-  Education & research
-  Economy & labor
-  Health & social well-being

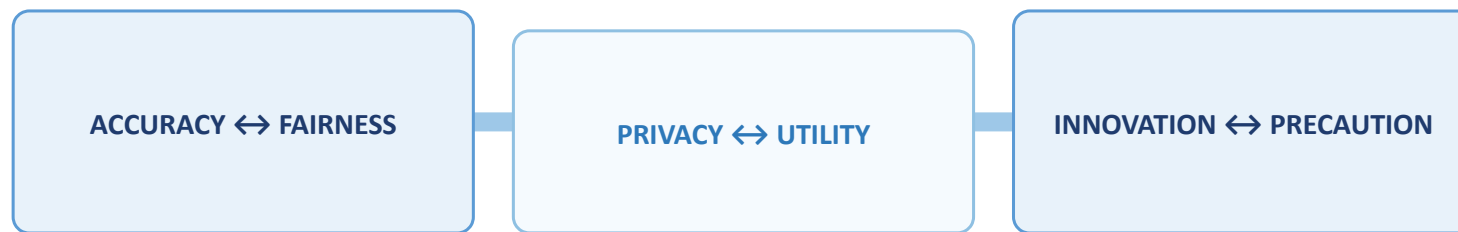


Ethics becomes actionable when values guide assessment, institutions, and sector-specific policy.

Common Tensions & Trade-offs

Principles rarely give a single answer.

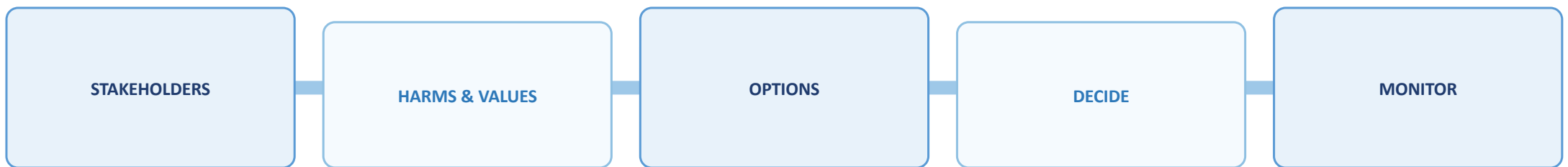
- Accuracy ↔ Fairness
- Privacy ↔ Utility
- Transparency ↔ IP / Security
- Innovation ↔ Precaution
- Automation ↔ Human dignity / oversight
- Trade-offs require judgment + context.



Trade-offs should be demonstrated, documented, and monitored — not assumed before better options are explored.

A Practical Ethical Decision-Making Framework

- **1. Identify** stakeholders & affected groups.
- **2. Map** values & potential harms (benefits, risks, rights).
- **3. Apply** ethical lenses (consequentialist, deontological, virtue, justice).
- **4. Generate & compare** options.
- **5. Decide, document & justify** (accountability).
- **6. Monitor & review** (iterative).



A defensible decision identifies stakeholders, maps harms, applies lenses, compares options, justifies action, and monitors outcomes.

Applying the Framework

Walkthrough: an AI hiring tool.

The case

- A tool that screens resumés and ranks applicants.

Stakeholders

- Applicants (esp. minorities), employer, society.

Harms / values

- Bias, privacy, fairness, efficiency.

Applying the 6 steps

- **3–** Lenses: justice; non-maleficence.
- **4.** Options: deploy / audit-first / human-in-loop / don't deploy.
- **5.** Decision: audit + human review + document.
- **6.** Monitor: adverse-impact ratios; retrain; review.



For a hiring tool, audit and redesign before deployment is often more defensible than deploying first and repairing harm later.

Capstone Exercise: Ethical Review of an AI Hiring Tool

Work in groups: 10 minutes analysis + 5 minutes debrief.

1 Stakeholders

2 Evidence needed

3 Benefits & harms

4 Ethical lenses

5 Options

6 Decision

7 Monitoring

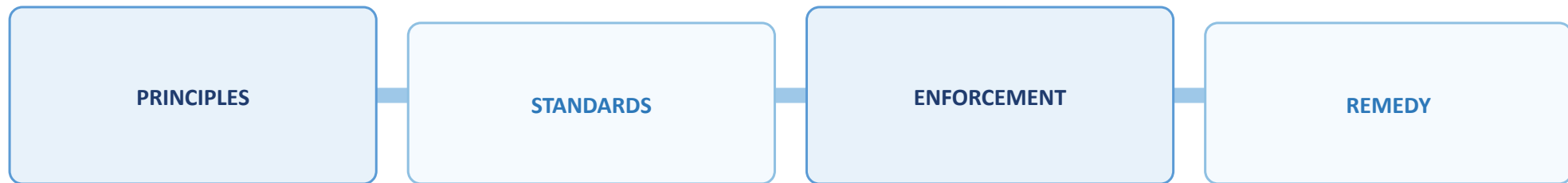
8 Remedy

Deliverable: a two-sentence recommendation, the evidence that could change it, and one safeguard for affected applicants.

A defensible ethical review links values to evidence, options, accountability, and remedy.

Limits of Principles & the Need for Governance

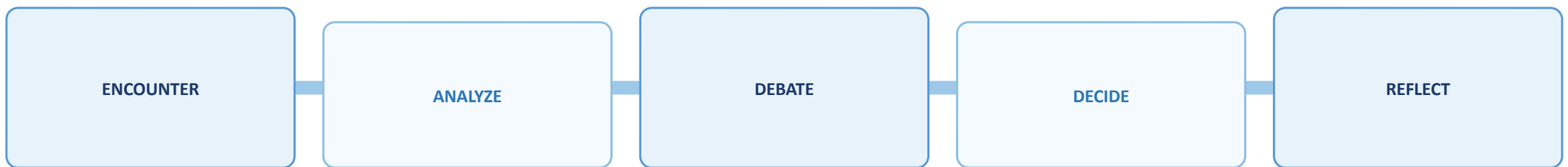
- Principles are necessary but **not sufficient**.
- They can't resolve hard trade-offs or prevent **"ethics washing"**.
- **Need governance & regulation:** audits, liability, standards, oversight.
- "Ethics is the floor; **law / governance is the enforceable structure.**"
- **Preview:** *Week 11 — AI Policy & Regulations.*



Principles set direction; governance assigns authority, incentives, evidence requirements, enforcement, and remedy.

How This Course Approaches AI Ethics

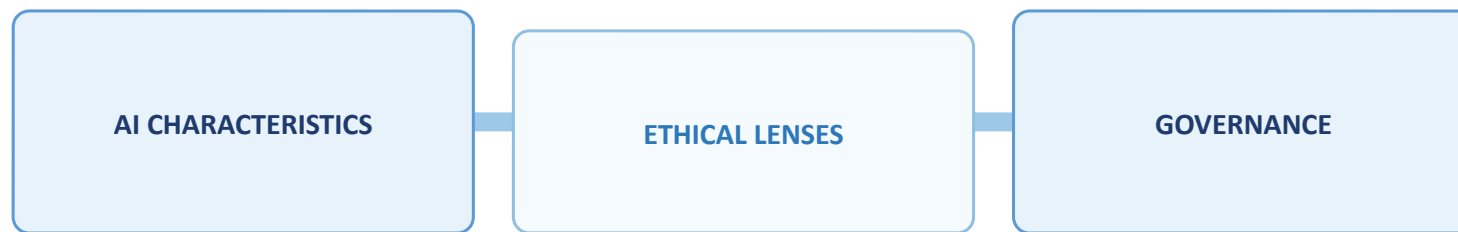
- **Case-based learning:** real cases ground abstract principles.
- **Weekly readings:** the five books + Canvas resources.
- **Active participation:** discussions, quizzes, AI news presentations.
- **Capstone:** final project — report + presentation.
- Each upcoming week deep-dives one theme from today's map.



The course repeatedly practices a cycle: encounter a case, analyze it, debate options, decide, and reflect.

Key Takeaways

- AI is **powerful + opaque + pervasive** → ethics is essential.
- Multiple ethical lenses illuminate **different facets** of a problem.
- Principles converge, but **implementation is hard** and governance matters.
- Real cases (O’Neil, Eubanks) show harms fall hardest on **the vulnerable**.
- You will build your own **ethical-reasoning toolkit** over the term.



Responsible AI requires both ethical reasoning and institutional mechanisms that make responsibility real.

Reading, Next Lecture & Q&A

- **Read:** Coeckelbergh (intro) + O'Neil Ch.1 (released on Canvas).
- **Next:** Lecture 2 — Privacy and Data Protection.
- **Reflect:** “Name an AI system you use daily — what ethical questions does it raise?”
- **Thank you! Questions are welcome.**

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Thank you!
Q&A